Jaadari Mohamed Amine

Telecommunications Engineering Student | AI Focus

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PROFILE

Telecommunications engineering student at the National Engineering School of Tunis (ENIT), specializing in Data Science, with hands-on experience in machine learning, multi-agent systems, and large-scale data analysis. Strong skills in AI, reinforcement learning, and RAG systems. Seeking a research internship starting in February 2027 to design scalable, intelligent, and high-impact solutions.

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Python, JavaScript, C#
- **Data Science & Analysis:** Data Exploration & Cleaning, Statistical Analysis, Feature Engineering, Data Visualization (Matplotlib, Seaborn, Plotly), Data Structures (Pandas DataFrames, NumPy arrays)
- **Machine Learning / AI:** Supervised/Unsupervised Learning, Deep Learning, Reinforcement Learning, RAG Systems, LLM Integration (LangChain)
- **Signal Processing:** EEG Signal Analysis, Feature Extraction, Filtering, Medical Data Processing
- **Databases:** MySQL, SQL Server, MongoDB, NoSQL
- **Libraries:** PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib
- **Tools / DevOps:** Git, Docker, Linux, MLOps
- **Certifications:** Google – Agent Fundamentals (DataCamp); Building Scalable Agentic Systems (DataCamp)

PROFESSIONAL EXPERIENCE

Vermeg for Banking Solutions — Software Engineering Intern *June 2026 – July 2026*

- Designed and developed a multi-agent system (Python, LangChain) integrating the Llama 3.3-70B LLM (Groq) for automated generation of personalized reports.
- Deployed a secure RESTful API for external integration, with automated multichannel distribution (Email, WhatsApp) and CSV database management.

PROJECTS

Automated EEG Analysis and Classification System with RAG Assistance *Jan. 2025 – June 2025*

- Analyzed and cleaned raw EEG time-series data by handling missing values, detecting outliers, and preprocessing signals; organized and manipulated the data using Pandas DataFrames and NumPy arrays.
- Visualized EEG signals through time-series plots, spectrograms, and correlation matrices to identify patterns and validate engineered and extracted features.
- Developed an end-to-end Alzheimer's disease classification system, achieving 81% accuracy on **real patient data** while comparing KNN, SVM, Random Forest, and Decision Tree models.
- Integrated a RAG pipeline for natural-language querying of medical reports and automated PDF report generation.

Reinforcement Learning Trading Bot *October 2026*

- Developed an RL agent (Monte Carlo Control) integrating an epsilon-greedy policy under GLIE constraints to balance exploration and exploitation, with iterative Q-table updates.

Big Data Analysis with Hadoop *March 2025*

- Research project on the Hadoop ecosystem (MapReduce, Hive) for big data analysis.

Facial Recognition System — ENIT *March 2025*

- Developed a real-time facial recognition system integrating Arduino hardware with Python/OpenCV.

EDUCATION

National Engineering School of Tunis (ENIT) *Sep. 2024 – Present*
Telecommunications Engineering (Data Science Specialization)

- Three-year curriculum focused on computer science, telecommunications systems, and software development life cycles.

MP Preparatory Classes — IPEI Gabès *2022 – 2024*

- Built a strong foundation in mathematical analysis, linear algebra, mechanics, and problem-solving, with an introduction to algorithms and programming.

LEADERSHIP & ACTIVITIES

- **IEEE CS Chapter (ENIT)** — Member; participated in computer science conferences and training sessions.
- **G2FOSS ENIT** — Training in algorithmic problem-solving and competition preparation (since Oct. 2024).
- **IEEE SIGHT Chapter (ENIT)** — Created visual content, managed social media, and strengthened the club's digital presence with the IT team.